Airbnb Business Analysis Report

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1. Business Understanding and Business Question

Airbnb prices vary with property and location characteristics (Gibbs *et al.*, 2018; Wang and Nicolau, 2017), while segmentation can reveal actionable submarkets (Dolnicar, 2020). We therefore ask whether Airbnb could benefit from higher prices in selected New York City segments by examining how advertised price relates to review-based demand and identifying the strongest pricing opportunities.

2. Data Understanding

2.1 Data Preparation

We used the 2019 Inside Airbnb snapshot of New York City (Gomonov, 2019; Appendix A), keeping listings with a positive price, minimum stay of at most 365 days and positive availability: 31,340 listings (Table 1). Listings open under 30 days are unreliable for demand, leaving 26,175.

TABLE 1 | Evolution of records

Preprocessing step	Removed	N remaining	Reason
Total records	—	48,895	Kaggle export of Inside Airbnb NYC 2019
Drop price = 0	11	48,884	Entry errors
Drop minimum_nights > 365	14	48,870	Entry errors
Drop availability_365 = 0	17,530	31,340	Clients cannot post reviews without bookings
Flag availability_365 < 30 (kept)	5,165 flagged	31,340	Reviews per available day inflated by seasonality; demand_reliable = 0
Modelling dataset (segmentation, regression)	—	31,340	
Demand subset	—	26,175	demand_reliable = 1; demand proxy capped at the 99th percentile

2.2 Feature Engineering

Research explains accommodation prices through listing characteristics and location (Rosen, 1974; Gibbs *et al.*, 2018; Deboosere *et al.*, 2019); with no booking data, several variables were engineered.

TABLE 2 | Engineered features

Feature	Description
dist_center_km	Distance in kilometres from the listing to Times Square, computed from the coordinates. Distance to the city centre is the location variable the pricing literature finds most important (Wang and Nicolau, 2017; Gibbs *et al.*, 2018; Gyódi and Nawaro, 2021). It is standardised before clustering.
demand_rate_year	A review-based demand proxy accounting for differences in listing availability. The formula: reviews_per_month × 12 / (availability_365 / 365). The number of reviews a listing would collect in a year if it were open all year. Inside Airbnb (n.d.) uses review rates as its occupancy proxy because bookings are not public. They assume 50% of all bookings leave a review. Listings open fewer than 30 days are flagged as unreliable and excluded from the demand analysis, and the proxy is capped at the 99th percentile. Refer to **table 1**.The proxy does not account for minimum nights: a guest reviews once per stay, so a listing with a 30-night minimum collects far fewer reviews than a 3-night listing at the same occupancy, and its demand is understated. We therefore control for minimum nights in the demand models; the demand levels themselves remain uncorrected.
availability_bucket	The host's availability regime, from availability_365: occasional (under 30 days), seasonal (30-120), half-year (121-240), most of the year (241-330) and always on (331-365).
stay_regime	The minimum-stay regime, from minimum_nights: short (up to 7 nights), mid (8-29) and long-term (30 or more). The 30-night line is New York's legal boundary for short-term rentals (Schaal, 2024).
log_price	log10 of the advertised nightly price. Price is heavily right-skewed, and the log stabilises variance.

2.3 Descriptive Statistics

- We use log because advertised price is heavily right-skewed (median $112, mean $162, maximum $10,000). Log is significantly more symmetric (Figure 1).
- Kruskal–Wallis tests showed significant differences across all groupings (all $p < 0.001$; Kruskal and Wallis, 1952), so effect sizes assessed practical importance (Table 3). Room type explained 46% of price-rank variance and distance 21%; neighbourhood explained 31% alone but only 6% beyond these factors, while all others explained 2% or less. Thus, room type and location define market structure, consistent with prior Airbnb pricing studies (Wang and Nicolau, 2017; *Gibbs et al.*, 2018; Deboosere *et al.*, 2019).

- The correlation matrix (Figure 2) confirms this and shows that minimum_nights correlates at −0.38 with the demand proxy because a guest reviews once per stay regardless of length. We will control for it.

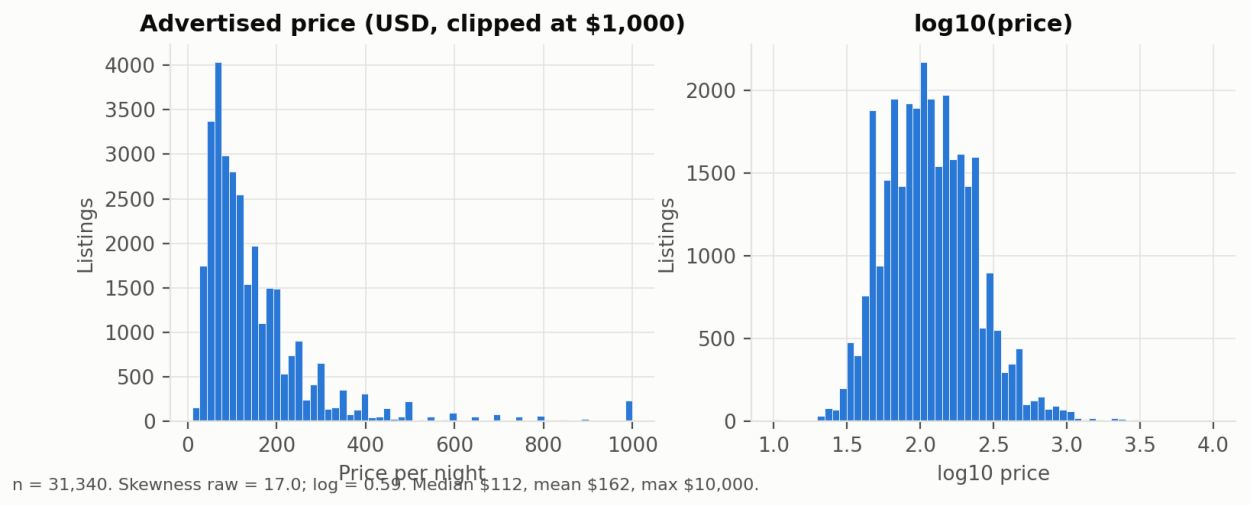


FIGURE 1 | Advertised price is heavily right-skewed; log10(price) is close to symmetric and is our modelling target.

TABLE 3 | Kruskal-Wallis tests and effect sizes, alone and after controlling for room type and distance

Grouping	Groups	Kruskal-Wallis H	p	Effect size alone (ε^2)	After controlling room type + distance (ΔR^2)
Room type	3	14,476	< 0.001	0.46	—
Distance band (< 4 / 4-8 / 8-14 / > 14 km)	4	6,456	< 0.001	0.21	—
Neighbourhood	218	10,012	< 0.001	0.31	+0.063
Borough	5	5,088	< 0.001	0.16	+0.013
Professional host (≥ 3 listings)	2	396	< 0.001	0.013	+0.0001
Has reviews	2	333	< 0.001	0.011	+0.008
Availability bucket	5	153	< 0.001	0.005	+0.014
Stay regime	3	112	< 0.001	0.004	+0.011

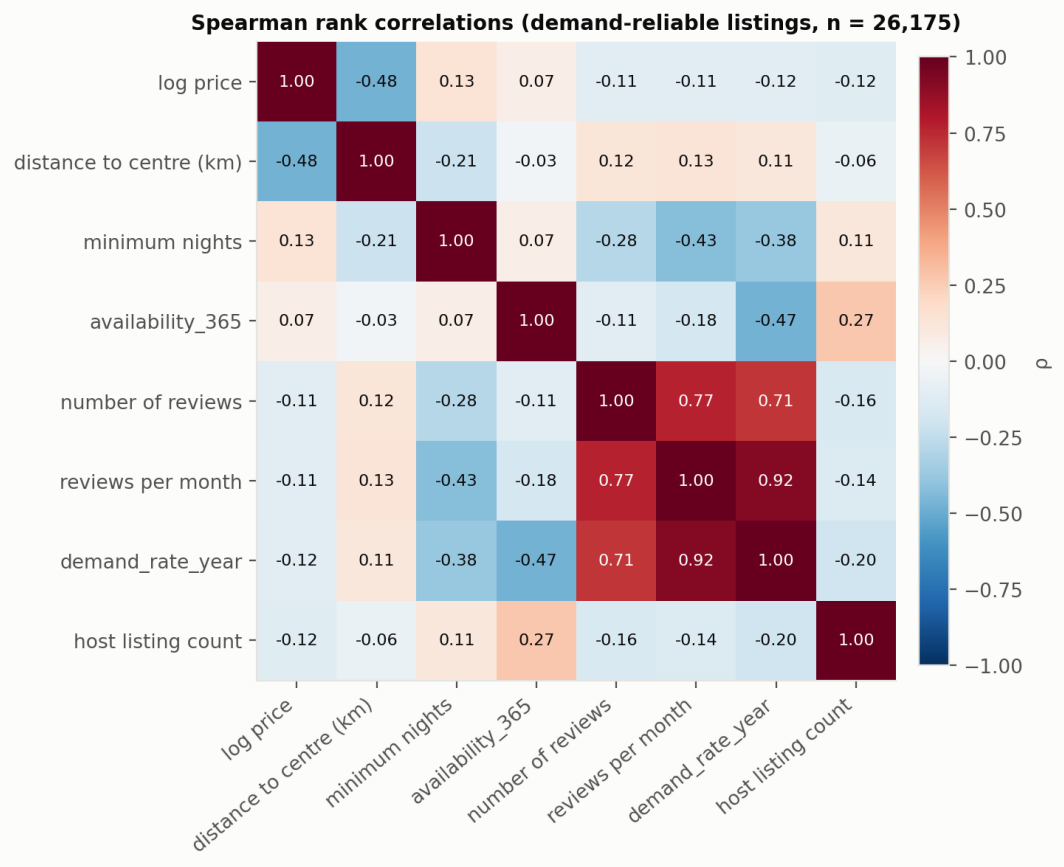


FIGURE 2 | Spearman rank correlations between the variables used, on the demand-reliable listings (n = 26,175). Log price relates to distance (−0.48) and little else; the demand proxy relates to minimum nights (−0.38) and, by design, to availability (−0.47).

3. Methodology

Three steps. First, K-prototypes clustering (Huang, 1998) segmented listings on room type and distance from the centre (k = 2-8 compared on silhouette, stability, size and interpretability; Rousseeuw, 1987), leaving price and demand out to avoid circularity. Second, price was predicted with a median benchmark, OLS, Elastic Net, Random Forest and CatBoost (80/20 split, five-fold cross-validation). Third, the demand proxy was related to price within each segment and market-wide, with an interaction test and robustness checks.

The code we created for data preparation, clustering, predictive modelling and robustness analysis is available in the project GitHub repository: https://github.com/ishalaan/airbnb-nyc-business-analysis/

4. Results and Findings

4.1 Listing Segments

The clustering confirms Table 3: adding availability and stay behaviour weakened separation (silhouette 0.660 to 0.406, two clusters), so room type and distance were kept. k = 5 had the highest silhouette (0.660, stability ARI 0.763); k = 8 was near-tied and more stable, and five clusters were retained as the balance of separation, market detail and interpretability (Table 4; Dolnicar, 2020). Shared rooms (2.7%) were absorbed into other clusters. γ = 1 was used; 0.75-2 gave the same partition.

TABLE 4 | Characteristics, pricing and demand patterns across the five listing segments

Segment	Key characteristics	Price pattern	Demand proxy
Cluster 1: Central entire homes	Entire properties within core Manhattan (n = 9,732); median 3.38 km to Midtown	Highest median rate ($199); strongest price resilience and demand retention at top quartiles	Lowest median demand proxy (13.05 reviews/year); trades volume for premium nightly margin
Cluster 2: Less-central entire homes	Entire homes/apartments in inner Brooklyn/Queens (n = 6,260); median 9.26 km to Midtown	Mid-to-premium median rate ($135); moderate price sensitivity at upper tiers	Highest median demand proxy across all segments (33.20 reviews/year); primary volume opportunity
Cluster 3: Central private rooms	Private rooms in accessible transit hubs (n = 5,887); median 4.82 km to Midtown	Moderate rate ($90); balanced trade-off between location convenience and shared space	Stable, consistent demand turnover (22.26 reviews/year)
Cluster 4: Less-central private rooms	Private rooms in outer boroughs (n = 6,917); median 9.46 km to Midtown	Low budget baseline ($60); highly compressed price dispersion	Solid review velocity (22.45 reviews/year); steep demand drop-offs when priced above median
Cluster 5: Outer private rooms	Peripheral private rooms (n = 2,544); median 17.59 km from Midtown	Budget outer-ring baseline ($72); lowest high-price retention	Moderate demand proxy (24.47 reviews/year); most price-sensitive segment (adjusted coefficient −1.039)

4.2 Price-Demand Patterns

Higher price was associated with lower demand ($\rho = -0.12$; Figure 2). Controlling for minimum nights, host size, distance, room type and borough, the market-wide log-log coefficient was −0.702, implying about 0.7% lower demand for a 1% higher price. Slopes differed significantly by segment (F-test $p = 3.55 \times 10^{-10}$; Figure 3) and remained robust to alternative demand caps, a 60-day availability threshold and short-stay listings (≤7 nights, n = 20,858). Central entire homes were least price-sensitive

(−0.481) and uniquely had higher top-quartile demand than bottom-quartile demand (retention 1.130); outer private rooms were most price-sensitive (−1.039), while less-central entire homes had the highest demand (33.20) but weak top-quartile retention (0.578; Table 6). This ordering is consistent with documented location premiums (Gibbs *et al.*, 2018; Deboosere *et al.*, 2019).

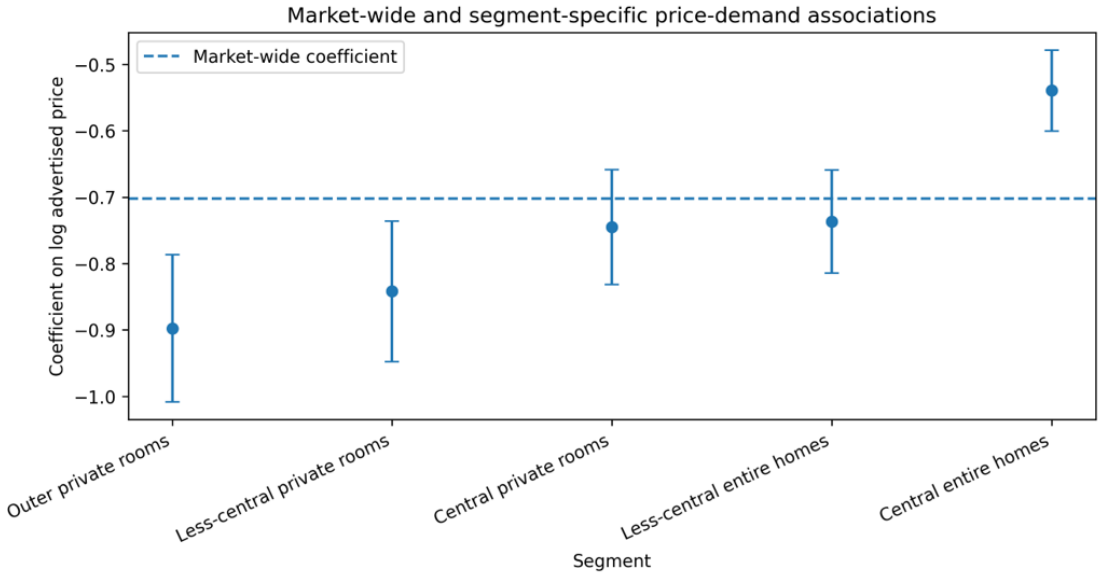


TABLE 6 | Within-segment price-demand relationship

Segment	Reliable N	Median price	Median demand proxy	Adjusted price coefficient (95% CI): % change in demand for a 1% higher price, after controls	Q4/Q1 demand retention: median demand in the top price quartile ÷ bottom quartile
Central entire homes	7,954	$199	13.05	−0.481 (−0.545 to −0.417)	1.130
Less-central entire homes	4,994	$135	33.20	−0.742 (−0.826 to −0.657)	0.578
Less-central private rooms	5,987	$60	22.45	−0.813 (−0.924 to −0.701)	0.727
Central private rooms	4,851	$90	22.26	−0.838 (−0.939 to −0.737)	0.899
Outer private rooms	2,389	$72	24.47	−1.039 (−1.160 to −0.917)	0.652
Market-wide (no segments)	26,175	$112	21.64	−0.702 (−0.741 to −0.664)	—

4.3 Regression Results

The regression predicted price from room type, distance, neighbourhood and borough plus minimum nights, availability, host listings and reviews per month; host listings stayed in because Deboosere *et al.* (2019) find professionalisation matters. CatBoost had the lowest cross-validation RMSE (0.1902), almost tied with Random Forest; both beat OLS and Elastic Net (Table 7, Figure 4). Room type was by far the most important predictor, then distance, as in Table 3.

TABLE 7 | Performance comparison of advertised-price prediction models

Model	CV RMSE (log10)	Test R^2 (log price)	Test MAE
Median benchmark	0.3134	≈ 0.000	$92.34
OLS	0.2085	0.5448	$66.82
Elastic Net	0.2088	0.5415	$66.91
Random Forest	0.1911	0.6196	$61.14
CatBoost (selected)	0.1902	0.6196	$61.89

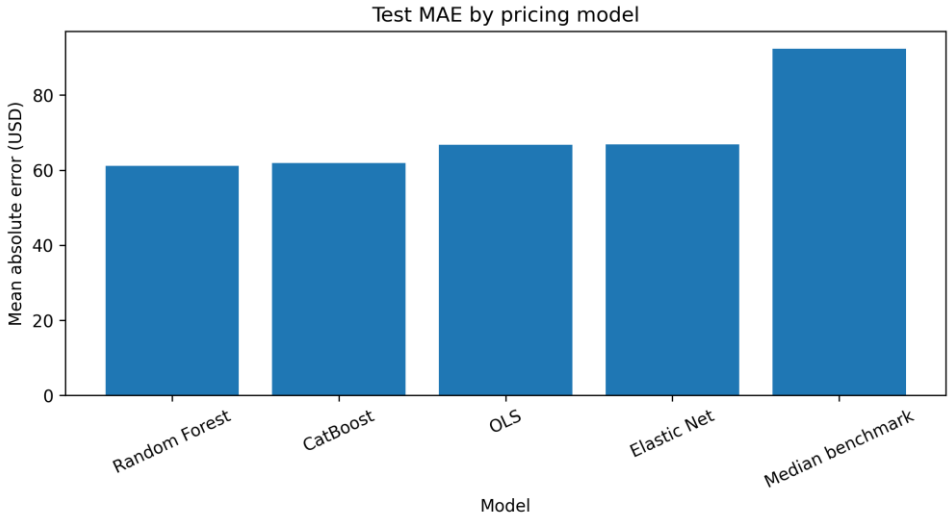


FIGURE 4 | Model comparison: test MAE by pricing model.

5. Business Insights and Recommendations

Higher advertised price goes with lower review-based demand across New York, but the strength differs by segment, so one rule does not fit the market (Table 8). Central entire homes are the strongest case for premium-price testing; they already command the highest price, yet their demand is least sensitive and holds in the top quartile. Less-central entire homes have the highest demand at a lower price but lose it at the top of

their range, here experimentation is required. Outer private rooms are the most price-sensitive and competitive.

TABLE 8 | Summary of business signals

Business signal	Segment	Evidence	Recommendation
Strongest price resilience	Central entire homes	Adjusted coefficient −0.481; Q4/Q1 retention 1.130; highest median price $199	Test selective premium price increases
Strongest demand opportunity	Less-central entire homes	Highest median demand 33.20 at $135; retention 0.578	Volume market; test prices carefully, no general increase
Most price-sensitive	Outer private rooms	Adjusted coefficient −1.039; retention 0.652	Maintain price competitiveness
Market-wide	All segments	Coefficient −0.702; interaction $p = 3.55 \times 10^{-10}$	Segment-specific pricing strategy

6. Limitations

- Demand is inferred from review volume, capturing about half of completed stays (Inside Airbnb, n.d.).
- Advertised rates exclude cleaning fees, commissions and seasonal discounts; there are no bookings, occupancy or revenue.
- The demand proxy ignores minimum nights (Table 2). Long stay listings collect fewer reviews at equal occupancy, so their demand is understated. It cannot be safely modelled: stay length is unobserved, and converting stays to nights with the minimum (Inside Airbnb, n.d.) assumes every stay equals it. This affects levels, not slopes: segments with many long stays, above all central entire homes, look

lower in demand than they are; the coefficients control for minimum nights and the short-stay check (Table 6) leaves the ranking unchanged.

- Shared rooms (861 listings) are too few to separate in cluster.
- The price model describes the market, not individual listings: its error is ten times larger in the top price quartile ($163 against $17), so it is unreliable for premium listings.

7. Conclusions

The analysis supports segment-specific rather than market-wide pricing. Central entire homes offer the strongest case for higher-price testing, while outer private rooms are more price-sensitive and less-central entire homes warrant controlled experimentation. However, because demand is proxied by reviews rather than bookings, occupancy or revenue, the findings indicate where Airbnb should test price increases rather than optimal listing-level prices.

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Appendix A: Description of the dataset

TABLE A1 | Data dictionary of the final dataset (n = 31,340 listings)

Field	Definition	Type	Min	Median	Mean	Max
id	Unique listing identifier	Identifier				
name	Listing title written by the host	Text				
host_id, host_name	Identifier and first name of the host	Identifier, text				
neighbourhood_group	Borough the listing is in	Categorical (5)				
neighbourhood	Neighbourhood the listing is in	Categorical (218)				
latitude	Latitude of the listing (anonymised by Airbnb to about 150 m)	Numeric	40.50	40.72	40.73	40.91
longitude	Longitude of the listing	Numeric	-74.24	-73.95	-73.95	-73.71
room_type	What is rented: the entire home/apartment, a private room, or a shared room	Categorical (3)				
price	Advertised nightly rate in USD at the time of the snapshot; the source does not say whether it is an average over the year, and it is not the price paid	Numeric	10	112	162	10,000
minimum_nights	Minimum number of nights a guest must book	Integer	1	3	8.0	365
number_of_reviews	Total reviews received over the life of the listing, as of the snapshot	Integer	0	10	31.9	629
last_review	Date of the most recent review; missing means never reviewed (17%)	Date				
reviews_per_month	Average reviews per month since the first review; missing set to 0 for never-reviewed listings	Numeric	0	0.92	1.51	58.5
calculated_host_listings_count	Number of NYC listings the same host operates	Integer	1	1	10.3	327

Field	Definition	Type	Min	Median	Mean	Max
availability_365	Days in the coming year on which the host has opened the calendar	Integer	1	167	176	365

Statistics are computed after the cleaning steps in Table 1. Field definitions follow Inside Airbnb (n.d.).

Appendix B: Robustness checks

TABLE B1 | Robustness checks: does the result survive under changing assumptions

Check	What was changed and why	Reliable N	Highest demand	Least price-sensitive (slope)	Most price-sensitive (slope)	Slopes differ by segment (p)
Main analysis	Demand proxy capped at the 99th percentile; listings with at least 30 available days	26,175	Less-central entire homes	Central entire homes (−0.54)	Outer private rooms (−0.90)	3.6×10^{-10}
Outliers (1)	No cap on the demand proxy, to see whether a few extreme review rates drive the result	26,175	Same	Same (−0.54)	Same (−0.91)	2.5×10^{-10}
Outliers (2)	Stricter cap at the 95th percentile	26,175	Same	Same (−0.53)	Same (−0.88)	4.5×10^{-10}
Availability threshold	Only listings with at least 60 available days, to remove short calendars whose review rates are inflated by season timing	23,163	Same	Same (−0.48)	Same (−0.82)	5.1×10^{-10}
Minimum nights	Only listings with a minimum stay of 7 nights or fewer, so that a review approximates a booking and the per-stay bias of Section 6 is largely removed	20,858	Same	Same (−0.69)	Central private rooms (−0.95); outer private rooms −0.94	4.2×10^{-5}